

Graduate School of Information Security, Doctoral Program Research (Preference) Proposal

Sajib Kumar Mitra

1. Research topic I would like to pursue in graduate school.

1-1. Research Theme and Background

I would like to conduct research on deployable hybrid quantum–classical edge intelligence for sustainable IoT security and anomaly detection.

Billions of Internet of Things (IoT) endpoints generate continuous telemetry that must be analyzed under strict energy, latency, and trust constraints. Sustainable intelligent IoT systems are judged by network-level outcomes—energy efficiency, resilience, cross-tier interoperability, and long operational lifetimes. Classical edge artificial intelligence (AI) already enables near-sensor inference and reduces backhaul, yet high-dimensional streaming workloads, adversarial security analytics, and post-quantum transition pressures motivate selective use of quantum artificial intelligence (Quantum AI) in hybrid pipelines rather than cloud-only variational models.

Quantum AI—the fusion of quantum computing with machine learning—is increasingly proposed for IoT security, kernel analytics, and combinatorial resource allocation. Contemporary QML toolchains such as PennyLane and Qiskit make hybrid quantum–classical learning pipelines practical for research, while edge gateways and FPGA SoCs create a realistic path toward compressed on-device inference. My ongoing systematic survey on Quantum AI-powered embedded devices for sustainable intelligent IoT systems (PRISMA-inspired) classifies IoT deployments into four paradigms—conventional, AI-based, quantum-enhanced, and quantum AI-based IoT—and shows that accuracy, circuit depth, algorithmic complexity, and deployable resource proxies are rarely reported jointly. This documented gap motivates doctoral research that co-designs hybrid models, benchmark protocols, and embedded prototypes under explicit power and latency budgets.

1-2. Research Objectives and Challenges

The rapid adoption of Internet of Things (IoT) systems in smart buildings, industrial environments, and healthcare has significantly expanded the attack surface of modern digital infrastructures. While Artificial Intelligence (AI)-based intrusion detection and anomaly detection techniques have become increasingly popular for identifying cyber threats, practical IoT deployments face two critical challenges. First, existing anomaly-detection systems often struggle to detect zero-day and stealthy attacks while operating under the strict energy, latency, and computational constraints of edge devices. Second, current research largely assumes that AI models, inference pipelines, and edge devices are inherently trustworthy, overlooking important security threats such as model theft, model tampering, firmware compromise, unauthorized device enrollment, and supply-chain attacks.

Recent advances in TinyML and lightweight edge AI have demonstrated promising deployment capabilities for constrained devices. However, these approaches primarily focus on reducing computational complexity and rarely address the protection of AI assets themselves. At the same time, emerging Quantum Machine Learning (QML) techniques have shown potential advantages in learning complex patterns and anomaly characteristics, but existing studies remain largely simulator-centric and provide limited evidence regarding deployment feasibility, security benefits, and practical operation in real-world IoT environments.

From a security perspective, practical IoT deployments require more than accurate anomaly detection. They require trusted execution of AI models, secure device identity management, secure boot processes, remote attestation, secure key management, and protection of deployed AI models from unauthorized modification or extraction. Therefore, an important research challenge is to integrate advanced AI-based detection mechanisms with trustworthy computing architectures capable of protecting both the device and the deployed intelligence. Four major barriers currently hinder the deployment of trustworthy AI-enabled IoT security systems.

Problem 1: Existing anomaly-detection research focuses primarily on detection accuracy while neglecting the protection of AI models, inference pipelines, and deployed security services.

Problem 2: Current Quantum AI and QML studies remain largely cloud- or simulator-centric and rarely evaluate practical deployment feasibility under realistic IoT edge constraints.

Problem 3: Existing studies lack unified evaluation frameworks that jointly measure detection performance, trusted-execution overhead, energy consumption, latency, and deployment complexity.

Problem 4: Most proposed solutions do not provide end-to-end architectures that integrate anomaly detection, trusted execution, secure model deployment, device authentication, remote attestation, and secure lifecycle management.

To address these challenges, this doctoral research aims to design, implement, and empirically validate a trustworthy hybrid quantum-classical edge-security framework for IoT systems. The proposed architecture combines anomaly detection with secure AI deployment mechanisms through Trusted Execution Environments (TEE), Hardware Root of Trust (HROt), secure boot, secure key management, and remote attestation. The research will investigate whether hybrid quantum-classical learning techniques can improve the detection of zero-day and stealthy attacks while maintaining practical energy, latency, and deployment constraints on resource-limited IoT edge devices. The research will investigate the following questions:

RQ1: Can hybrid quantum-classical models improve the detection of zero-day and stealthy IoT attacks compared with state-of-the-art lightweight classical AI models under fixed edge-computing constraints?

RQ2: How can Trusted Execution Environments, Hardware Root of Trust, secure boot, remote attestation, and secure key management protect deployed AI models and inference pipelines from tampering, theft, and unauthorized modification?

RQ3: What are the security, latency, energy, and deployment trade-offs when integrating hybrid Quantum AI with trusted edge-computing architectures on FPGA- and ARM-based IoT platforms?

2. Aspirations

My motivation for doctoral study stems from a desire to provide the world with highly secure and reliable information and communication systems, thereby contributing to a convenient and safe intelligent society. During my B.Sc. and M.S. studies in Computer Science and Engineering at the University of Dhaka, I became fascinated by the breadth and depth of security, quantum computing, and embedded systems. My M.S. thesis on efficient reversible logic and quantum circuit design, together with subsequent IEEE/ACM conference publications, established a research foundation in low-power quantum-classical co-design.

As Joint Director (ICT) at Bangladesh Bank, I have gained extensive experience in secure ICT architecture, software engineering leadership, and national-scale systems relevant to information-system audit and trustworthy operations. The field of hybrid Quantum AI for IoT is a convergence of machine learning, quantum computing, cryptography, and embedded deployment; conceptual design, metric alignment, and reproducible evaluation are therefore crucial. I believe that my combination of academic research in reversible/quantum computing, industry-scale ICT practice, and co-investigator experience on a systematic IoT–Quantum AI survey will be an advantage in advancing this doctoral research. Through the program, I aim to cultivate research execution ability on hardware-aware QML pipelines and technical leadership that bridges academic research and technological development.

3. Research Plan

3-1. Research Approach

This research adopts a security-driven architecture rather than a purely AI-driven approach. The proposed framework combines hybrid quantum-classical anomaly detection with trusted execution and secure deployment technologies to provide both effective attack detection and trustworthy operation in IoT environments.

The AI component will investigate Variational Quantum Circuits (VQC), Quantum Kernel methods, and hybrid quantum-classical learning architectures implemented through Qiskit and PennyLane. These techniques will be evaluated against classical machine learning and TinyML baselines to determine whether Quantum AI provides measurable advantages for detecting previously unseen attacks under practical edge-computing constraints.

The security component will focus on protecting deployed AI assets and IoT devices throughout their operational lifecycle. The framework will incorporate Trusted Execution Environments (TEE), Hardware Root of Trust (HROt), secure boot, secure key management, remote attestation, and trusted inference mechanisms. These technologies will ensure that AI models remain protected from unauthorized access, modification, extraction, and execution in compromised environments.

The primary experimental platform will consist of ARM-based IoT gateways or Raspberry Pi-class edge devices supporting trusted execution technologies such as ARM TrustZone. FPGA acceleration using Xilinx Zynq or Artix-class platforms will be investigated where appropriate to evaluate performance, energy efficiency, and hardware-assisted security features. Quantum model development and training will be conducted using Qiskit and PennyLane on simulators and available cloud quantum processors, while deployment experiments will be performed on physical edge devices.

The relationship between Quantum AI and trustworthy IoT deployment remains central to this research. Quantum computing components will only be incorporated when they demonstrate measurable benefits compared with classical alternatives. Consequently, all proposed approaches will be evaluated using unified metrics that simultaneously consider security effectiveness, detection performance, latency, energy consumption, memory requirements, trusted-computing overhead, and deployment feasibility.

3-2. Steps in the Research Process

This research will proceed through a series of iterative phases. Results will be disseminated through international conferences and journals, including security venues such as ACSAC, RAID, ESORICS, DIMVA, AsiaCCS, and related journals.

A. Security Model, Threat Analysis, and Benchmark Framework

The first phase focuses on establishing the security model, threat landscape, and benchmark framework. Representative use cases will include:

- Smart-building IoT security gateways
- Industrial IoT (IIoT) security systems
- Internet of Medical Things (IoMT) monitoring platforms

The threat model will include:

- Zero-day malware attacks
- Advanced persistent threats (APT)
- Lateral movement attacks
- Firmware tampering
- AI model extraction attacks
- AI model poisoning attacks
- Unauthorized device enrollment
- Supply-chain attacks

Classical and hybrid baselines will be established using public datasets and reproducible evaluation procedures. The outcome of this phase will be:

- Security requirements specification
- Threat model definition
- Benchmark protocol
- Initial classical AI baselines
- Trusted-computing architecture design

B. Hybrid Quantum-Classical Security Model Development

The second phase focuses on developing and evaluating hybrid quantum-classical anomaly-detection models. Candidate approaches include:

- Variational Quantum Circuits (VQC)
- Quantum Kernel Methods
- Hybrid Quantum Neural Networks
- Quantum-inspired classical models

The investigation will determine whether these methods improve detection performance against previously unseen attacks while maintaining practical deployment requirements. To support

deployment on resource-constrained edge platforms following Model-compression techniques will be explored.

- Circuit-depth reduction
- Angle encoding optimization
- Knowledge distillation
- Parameter pruning

C. Trusted AI Deployment Architecture

The third phase focuses on secure deployment and trustworthy execution. The proposed framework will integrate:

- Trusted Execution Environments (TEE)
- Hardware Root of Trust (HROt)
- Secure Boot
- Remote Attestation
- Secure Key Management
- Trusted Model Storage
- Trusted Inference Services

The objective is to protect AI assets and security services from tampering, theft, and unauthorized modification.

4. Career paths after Doctoral

Technological innovation in IoT, edge AI, and quantum computing is accelerating, and in order to lead security and embedded-intelligence R&D in the future, I feel it is necessary to solidify my own area of expertise and to manage highly specialized research activities from a technical standpoint. After obtaining my degree, I aim to establish expertise as a researcher in deployable hybrid edge intelligence for IoT security.

I also intend to leverage the research skills cultivated during doctoral study in corporate R&D management and technical guidance, becoming a person who can bridge the gap between academic research and technological development. Through these activities, I hope to collaborate with supervisors, FPGA and IoT testbed partners, and industry stakeholders to create synergies between foundational survey insights and reproducible prototypes, thereby increasing my contribution toward social implementation. Ultimately, I aspire to co-design trustworthy, energy-efficient IoT security analytics and to propose the architectural foundations needed for a convenient and secure intelligent society.